



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Wednesday, 8 October 2025, 10:04 AM**State** Finished**Completed on** Wednesday, 8 October 2025, 10:26 AM**Time taken** 21 mins 37 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:**Input:** 6**Output:** 6**Explanation:** There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      long long d[n+1];
6      for(int i=0;i<=n;i++){
7          d[i]=0;
8      }
9      d[0]=1;
10     for(int i=1;i<=n;i++){
11         d[i]=d[i-1];
12         if(i>=3){
13             d[i]+=d[i-3];
14         }
15     }
16     printf("%lld",d[n]);
17     return 0;
18 }
```

	Input	Expected	Got	
✓	6	6	6	✓

	Input	Expected	Got	
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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